Evaluating Advanced Machine Learning Models for Histopathological Diagnosis of Hansen Disease

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Introduction: Leprosy is a neglected infectious disease caused by *Mycobacterium leprae* and *Mycobacterium lepromatosis* and remains a public health challenge in tropical regions. Therefore, the development of technological tools such as machine learning (ML) offers an opportunity to innovate strategies for improving the diagnosis of this complex disease.

Objective: To validate the utility of different ML models for the histopathological diagnosis of Hansen disease.

Methodology: An observational study was conducted where 55 H&E-stained tissue slides from leprosy patients and 51 healthy skin controls were analyzed using microphotographs captured at various magnifications. These images were categorized based on histopathological findings and processed using the Cross-Industry Standard Process for Data Mining methodology for designing and training ML models. Five types of ML models were evaluated using standard metrics such as accuracy, sensitivity, and specificity, emphasizing data normalization as a fundamental step in optimizing model performance.

Results: The artificial neural network (ANN) model demonstrated an accuracy of 70%, sensitivity of 74%, and specificity of 65%, demonstrating its ability to identify leprosy cases with moderate precision. The receiver operating characteristic curve of the ANN model showed an area under the curve of 0.71. Conversely, models such as decision trees, logistic regression, and random forests showed similar accuracy results but with slightly lower sensitivity, potentially indicating a higher risk of false negatives in detecting leprosy-positive cases.

Conclusion: The ANN model emerges as a promising alternative for leprosy detection. However, further refinement of these models is necessary to enhance their adaptability across different clinical settings and participation in patient care.

Key Words: pathology, artificial intelligence, leprosy, algorithms, Mycobacterium

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INTRODUCTION

Hansen disease or leprosy is a neglected infectious disease caused by *Mycobacterium leprae* and *Mycobacterium lepromatosis*, which have a strong affinity for the skin and peripheral nerves.[1] These mycobacteria are responsible for disabilities affecting the eyes, hands, and feet. Historically, Colombia has been in a stage of postelimination for this disease since 1997; however, nearly 3 decades later, the country continues to report between 300 and 400 leprosy cases annually, including in children, indicating ongoing transmission of this infection in this Latin American country.[2,3]

The elimination of leprosy would not only improve the quality of life for leprosy patients but also promote social inclusion and reduce the economic burden associated with the disease. In the 21st century, new technological tools such as machine learning (ML), a branch of artificial intelligence, offer an opportunity to develop innovative strategies to improve the diagnosis of this complex disease.[4] The incorporation of artificial intelligence models not only represents a technological innovation in early detection but also enriches the epidemiological understanding of leprosy, reducing uncertainty in histopathological evaluations that require significant expertise from pathologists.[5] Given that leprosy presents with different clinical spectra (indeterminate, tuberculoid, dimorphic or borderline, and lepromatous forms),[6] its diagnosis and understanding are challenging. Therefore, the aim of this study was to validate different ML models that could be useful for the histopathological diagnosis of Hansen disease.

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This research was conducted following the international ethical standards given by the World Health Organization and the Pan American Health Organization, supported by the Declaration of Helsinki, promulgated in 1964, and the statutes given at the national level by resolution number 008430 of 1993 of the Ministry of Health of Colombia, which regulates health studies. In addition, it was endorsed by the Instituto Colombiano de Medicina Tropical- Universidad CES, Human Research Ethics Board (act No 90 of March 14, 2024).

H. Serrano-Coll and N. Cardona-Castro designed the study. J. P. Ospina-Gómez interpreted the study's histopathological findings. H. Serrano-Coll and M. Vargas-Clavijo wrote the manuscript. M. Vargas-Clavijo performed the data analysis. J. P. Ospina-Gómez, N. Cardona-Castro conducted a critical review. This research was directed by H. Serrano-Coll and N. Cardona-Castro. All authors read and approved the manuscript.

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation, to any qualified researcher.

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TABLE 1. Description of the Histopathological Patterns

Description	
Indeterminate leprosy	Lymphoplasmacytic inflammation at the perivascular, periadnexal, and perineural level
	Inflammation covering vessels, annexes, and nerves
Tuberculoid spectrum: tuberculoid forms and borderline tuberculoid	Well-formed granuloma surrounding annexes or nerves
	Multinucleated cells (Langhans cells) + epithelioid cells
	Fragmented nerves, not visible in high magnification fields (×40)
Lepromatous spectrum: borderline lepromatous and lepromatous leprosy	Poor-formed granuloma with foamy macrophages (Virchow cells); this extends throughout the skin
	Perineurium shaped "onion skin-like"
	Fibrous nerves
Healthy skin	Epidermis and dermis without evidence of inflammation, nerves, glands, and hair follicles preserved

Reference for this table Nunzi et al.[6]

MATERIALS AND METHODS

Study Type and Sample Size

A descriptive, observational cross-sectional study was conducted on 106 slides with hematoxylin–eosin (H&E)-stained tissue sections from 55 leprosy patients and 51 healthy skin controls. These samples were selected at convenience from the sample bank with extended informed consent for use in future research, and healthy skin samples were collected from discarded tissue resulting from plastic surgery on volunteer individuals with no personal or family history of leprosy. Furthermore, the leprosy patient samples used in the study were confirmed through clinical, bacteriological, histopathological, and molecular diagnostics. This included an initial clinical diagnosis, supported by specific histopathological findings in H&E staining, and additional tests such as acid-fast staining and DNA testing to ensure the accuracy of the leprosy disease diagnosis.

The H&E-stained tissue sections were re-evaluated and reclassified by a dermatopathologist. A total of 640 microphotographs were captured, comprising 318 microphotographs of healthy skin and 322 with leprosy (137 indeterminate, 84 tuberculoid spectrum, and 101 lepromatous spectrum). These were stored in JPG format at 4X, 10X, 40X using a LEICA dm500 optical microscope camera, and each of these microphotographs was categorized using the histopathological findings described in Table 1 and Figure 1.[6]

Design and ML Algorithm Training

The design and training of the ML algorithm were based on the Cross-Industry Standard Process for Data Mining methodology,[7] following these phases:

1. Problem Understanding: Initial review of microphotographs.
2. Data Understanding: Exploration of datasets using histograms, heat maps, and correlation charts.
3. Data Preparation: Validation, cleaning, integration, and confirmation of microphotograph format.
4. Modeling: Exploration and experimentation with various ML models, from linear regressions to artificial neural networks (ANNs).
5. Model Evaluation: Performance measurement using accuracy, sensitivity, specificity, and confusion matrix.

Workflow and ML Model Analysis

The 640 labeled microphotographs were loaded and normalized to ensure pixel values were within a range of 0–255. This normalization was crucial for efficient processing of ML models. Subsequently, the data were split into 80% for training and 20% for validation. Five different ML models were then trained: ANNs, decision trees (DT), support vector machines (SVM), logistic regression (LR), and random forests (RF).

Each model was evaluated using the following parameters:

1. Accuracy: Proportion of correct predictions out of total predictions.[8]
2. Sensitivity (Recall): Model's ability to correctly identify positive samples. Crucial in ensuring cases of leprosy are not missed.[8]
3. Specificity: Model's ability to correctly identify negative samples. Indicates accuracy in identifying healthy cases.[8]

In addition, Receiver Operating Characteristic (ROC) curves and loss curves were generated to evaluate the model with the best performance. These analyses were conducted using Python packages.[9]

RESULTS

Description of Samples Evaluated

Histopathological examination revealed that 48.1% of samples consisted of healthy skin. Among the leprosy samples, 24.6% were categorized as indeterminate, 13.2% as tuberculoid spectrum, and 14.2% as lepromatous spectrum. Microphotographs collected indicated that 49.7% corresponded to healthy skin, with 21.4% indeterminate, 13.1% tuberculoid spectrum, and 15.8% lepromatous spectrum (Table 2).

Validation of Different ML Models

Regarding the validation of different ML models, the ANN model showed an accuracy of 70%, with a sensitivity of 74% and specificity of 65%. These results underscore its ability to accurately identify positive cases, particularly notable for its relatively high sensitivity, crucial to early disease detection (Fig. 2 and Table 3).

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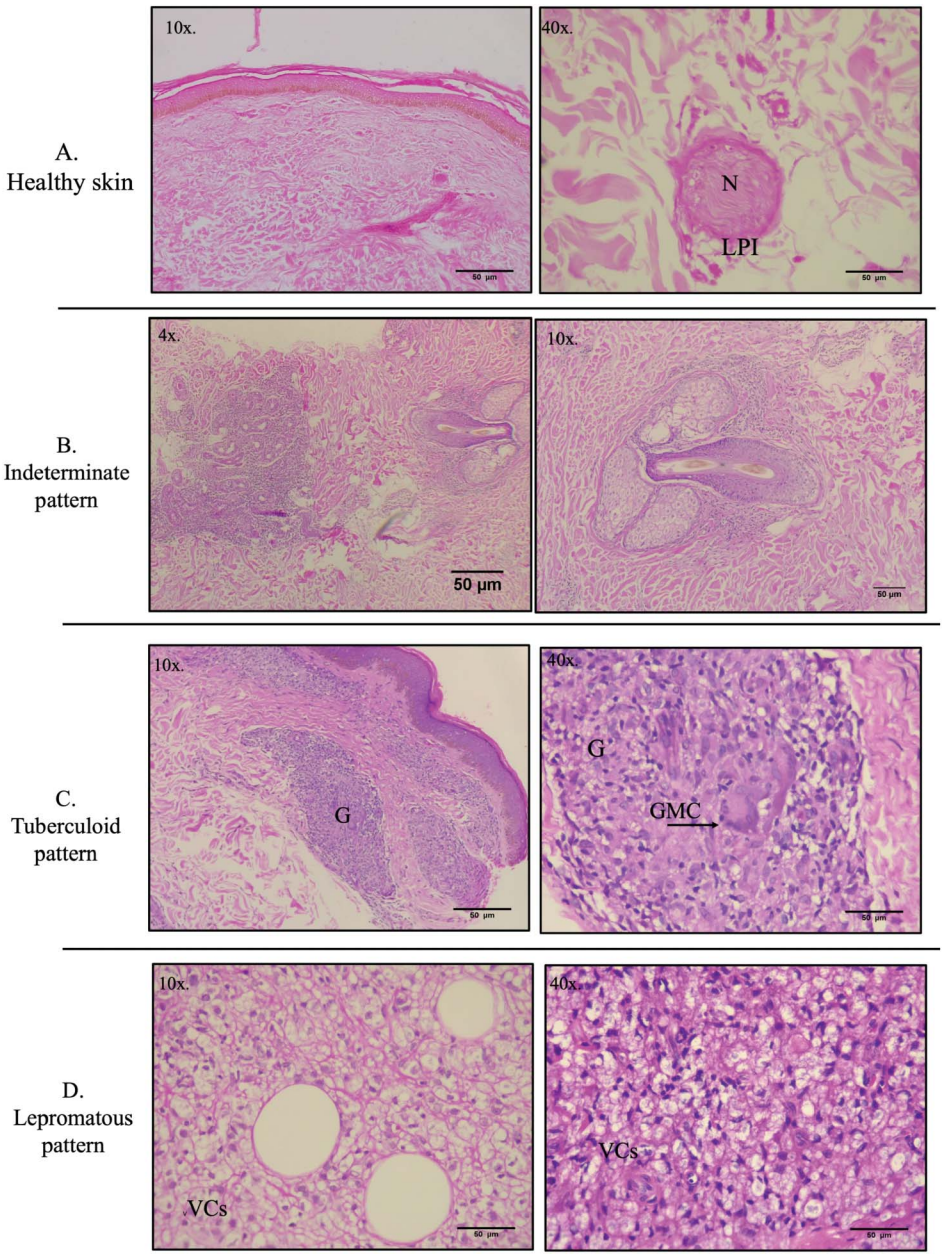


FIGURE 1. Microphotographs of the histopathological patterns evaluated. A, Microphotograph at ×10 magnification showing the epidermis and dermis without inflammatory signs; microphotograph at ×40 magnification shows a healthy dermal nerve without inflammatory changes. B, Microphotographs at ×4 and ×10 magnifications show a lymphoplasmacytic infiltrate at the dermal level. C, Microphotographs at ×10 and ×40 magnifications show well-formed granulomas (G) with presence of multinucleated giant cells (GMC). D, Microphotographs at ×10 and ×40 magnifications show Virchow cells with what are believed to be globi inside.

TABLE 2. Samples Evaluated and Microphotographs

Description	
Histopathological pattern n = 106 (100%)	
Indeterminate	26 (24.5)
Tuberculoid spectrum	14 (13.2)
Lepromatous spectrum	15 (14.2)
Healthy skin	51 (48.1)
Microphotographs n = 640 (100%)	
Indeterminate	137 (21.4)
Tuberculoid spectrum	84 (13.1)
Lepromatous spectrum	101 (15.8)
Healthy skin	318 (49.7)

On the other hand, other explored models—DT, LR, and RF—achieved accuracies of 67%, 65%, and 63%, sensitivities of 59%, 55%, and 63%, and specificities of 78% (DT, LR) and 71%, respectively (Figs. 3A–F and Table 3). These models demonstrated notable capability in correctly identifying negative cases, although their lower sensitivity might pose a higher risk of false negatives when analyzing leprosy-positive cases.

Finally, the SVM model exhibited the lowest accuracy at 61%, sensitivity of 47%, and highest specificity of 80% (Figs. 3G, H and Table 3). This model excelled in correctly identifying negative cases but showed limited sensitivity in detecting leprosy-positive cases, potentially leading to a higher risk of false negatives in clinical practice.

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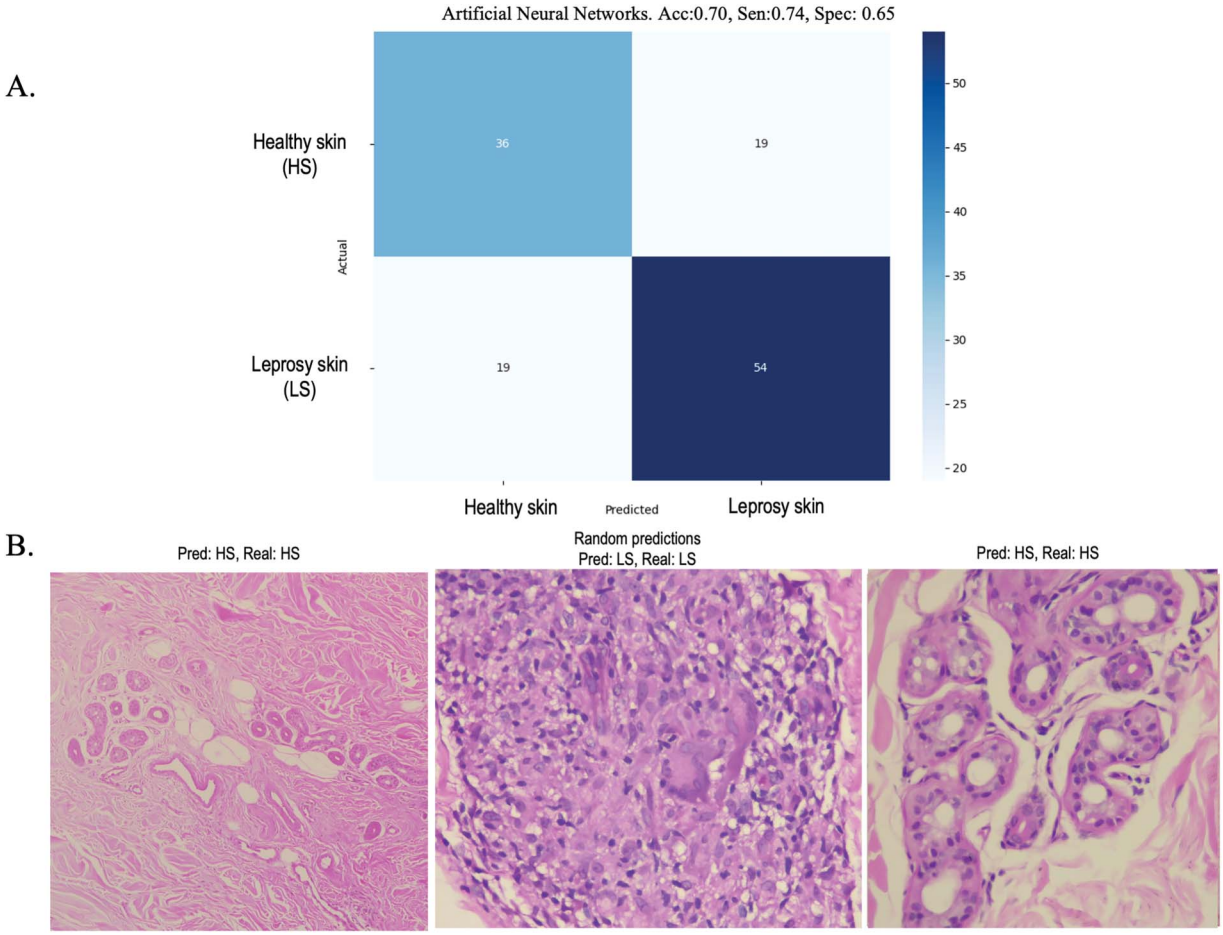


FIGURE 2. Confusion matrix and the prediction for the ANN model. A, The model correctly classified 36/55 healthy images and 54/73 leprosy images. The model accuracy is 0.70, sensitivity is 0.74, and specificity is 0.65. B, The images of healthy samples were correctly classified (pred: HS, real: HS). The image in the center shows a leprosy image correctly classified (pred: LS, real: LS). These images illustrate the model's ability to correctly identify both positive (leprosy) and negative (healthy) samples. HS, healthy skin; LS, leprosy skin. On the other hand, in the confusion matrix, darker or more intense colors along the main diagonal indicate a higher number of correct classifications, whereas darker colors off the diagonal signify errors, highlighting areas where the model may require improvement.

Performance of ANN Model via ROC and Loss Curves

The ROC curve provides a visual representation of the classification model's performance of true positive rate versus false positive rate at different classification thresholds.[10] The ROC curve for the ANN model lies above the diagonal line representing a random classifier, indicating a performance superior to that of random chance. The area under the curve (AUC) is 0.71, suggesting moderate performance in correctly classifying images. This AUC implies a 71% probability that the model correctly classifies a microphotograph showing active signs of this disease (Fig. 4A).

Furthermore, a loss curve was calculated to determine how quickly the model learns from the supplied data, observing a rapid decline in slope, indicating efficient parameter adjustment to minimize error. From approximately the 10th iteration, the curve stabilizes, approaching a very low value. This indicates that the model has reached a convergence point where additional iterations do not significantly improve its learning patterns (Fig. 4B).

TABLE 3. Evaluation of Different ML Models

Model	Accuracy	Sensitivity	Specificity
Artificial neural networks	0.70	0.74	0.65
Decision trees	0.67	0.59	0.78
Support vector machines	0.61	0.47	0.8
Logistic regression	0.65	0.55	0.78
Random forests	0.66	0.63	0.71

DISCUSSION

ANN models present 2 significant advantages over the other explored models. Their applicability to a wide variety of

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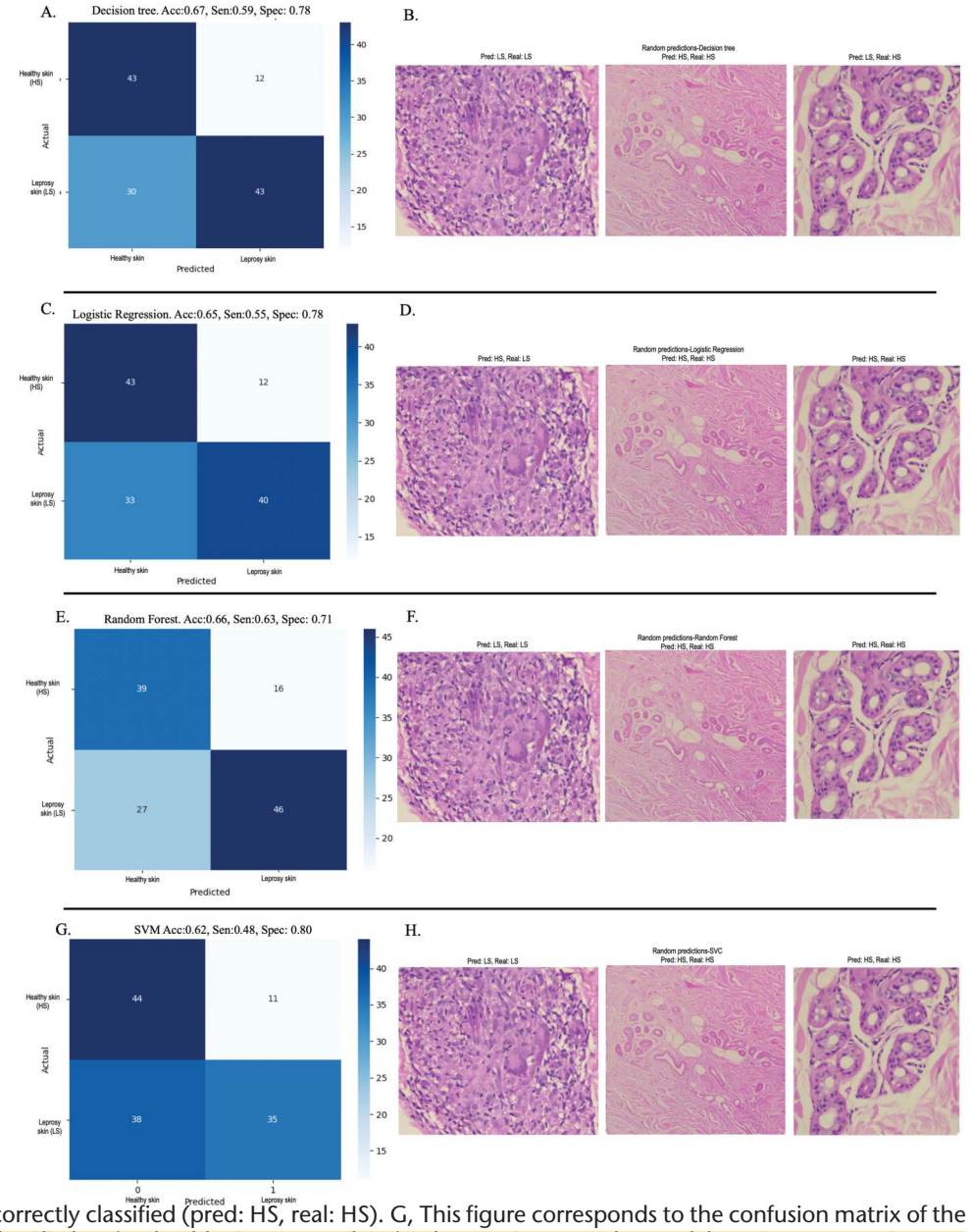


FIGURE 3. Confusion matrix and the prediction for the decision tree, LR, RF, and SVM models. A, This figure shows the confusion matrix of the decision tree model. The model correctly classified 43/55 healthy images and 43/73 leprosy images. The model accuracy (Acc) is 0.67, sensitivity (Sen) is 0.59, and specificity (Spec) is 0.78. B, The figure shows examples of predictions from the decision tree model on test images. The leftmost image shows a leprosy image correctly classified (pred: LS, real: LS), the center image shows a healthy image correctly classified (pred: HS, real: HS), and the image on the right shows a healthy image incorrectly classified as leprosy (pred: LS, real: HS). C, The figure shows the confusion matrix of the LR model, which correctly classified 43/55 healthy images and 40/73 leprosy images. The model Acc is 0.65, Sen is 0.55, and Spec is 0.78. D, The figure shows examples of predictions from the decision tree model on test images. The leftmost image shows a leprosy image incorrectly classified as healthy (pred: HS, real: LS). The center and right images show healthy images correctly classified (pred: HS, real: HS). E, This figure corresponds to the confusion matrix of the random forest model. The model correctly classified 39/55 healthy images and 46/73 leprosy images. The model Acc is 0.66, Sen is 0.63, and Spec is 0.71. F, Examples of predictions from the random forest model are shown. The leftmost image shows leprosy images correctly classified (pred: LS, real: LS). The center and right images show healthy images correctly classified (pred: HS, real: HS). G, This figure corresponds to the confusion matrix of the SVM model. The model correctly classified 44/55 healthy images and 35/73 leprosy images. The model Acc is 0.62, Sen is 0.48, and Spec is 0.80. H, This figure shows examples of predictions from the SVC model on test images. The leftmost image shows a leprosy image correctly classified (pred: LS, real: LS). The center and right images show healthy images correctly classified (pred: HS, real: HS). HS, healthy skin; LS, leprosy skin.

problems and relative ease of use. These advantages allow ANNs to be used in numerous fields, from pattern recognition to image processing and complex data classification. Their ability to learn and adapt to different data types and tasks makes them extremely versatile.[11] ANNs learn patterns from similar events and make decisions based on known interpretations, facilitating implementation due to the availability of numerous libraries and software tools that simplify their design and training.[12]

In the field of dermatopathology, ANNs have proven to be effective tools in image analysis due to their ability to recognize, process, classify, and segment images with high accuracy.[13] This ML model is used to differentiate between malignant and benign lesions, improve diagnostic accuracy, and predict recurrence of metastasis and disease-specific survival rates, such as in melanoma.[13] Moreover, their ability to handle large volumes of digital image data can enhance workflow efficiency and diagnostic accuracy in clinical settings. As technology advances and higher-quality data are collected, neural networks could revolutionize dermatopathology, offering faster, more accurate, and accessible diagnoses, even in resource-limited areas.[13]

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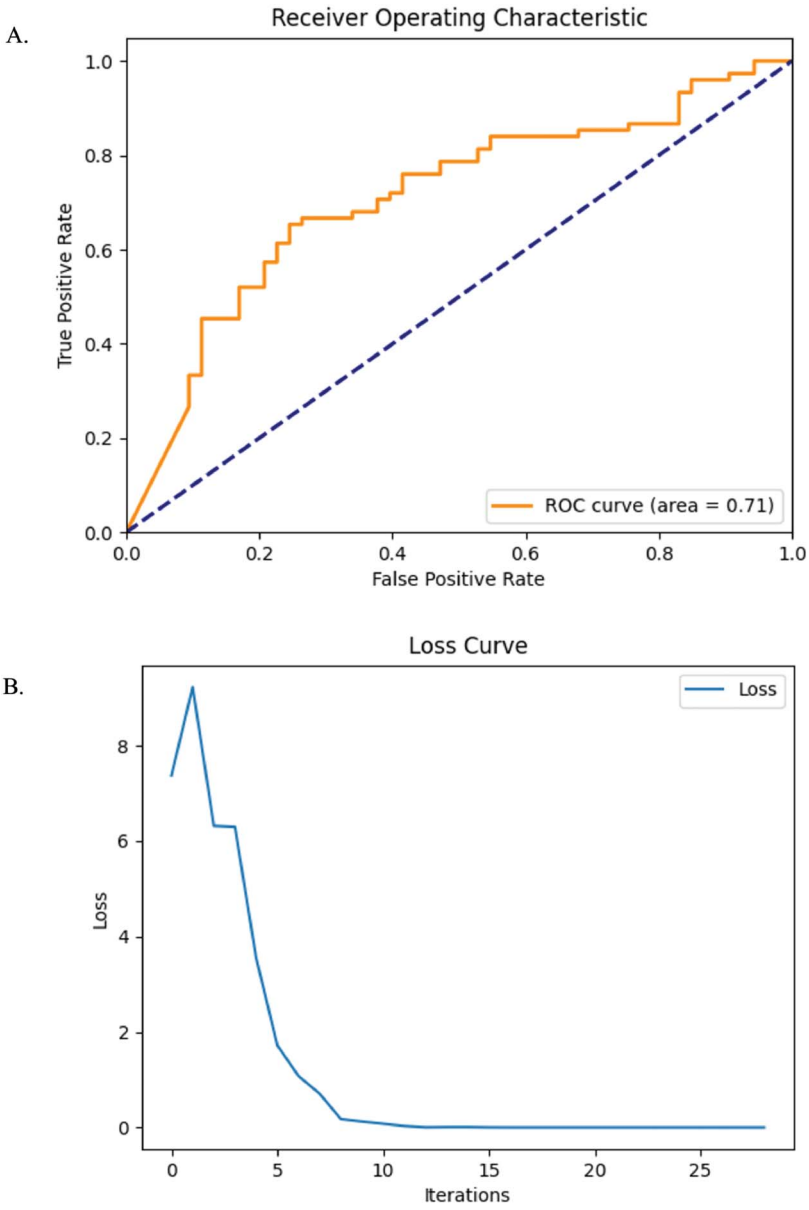


FIGURE 4. Performance of ANN model via ROC and loss curves. A, The ROC curve illustrates the model's performance of true positive rate versus false positive rate. An AUC of 0.71 indicates moderate performance in classification. B, The loss curve shows the decrease in error during model training. The *x*-axis represents the number of iterations, and the *y*-axis represents the loss. The initial rapid decrease followed by stabilization indicates that the model has converged.

However, the difficulties experienced by all evaluated models in distinguishing between leprosy-affected samples and healthy skin may partly be attributed to a percentage of samples classified as indeterminate leprosy. This early form of the disease is characterized by mild inflammatory involvement around cutaneous appendages or dermal nerves, which can resemble nonspecific skin inflammations such as contact dermatitis or degenerative skin changes related to aging or sun exposure.[14] Furthermore, the sensitivity of histopathological detection by expert pathologists for this clinical form is 69%, demonstrating slightly greater accuracy of these learning models compared with the trained human eye in making these complex leprosy diagnoses.[15]

The main limitation of this study is the need for more extensive training of ANN models using a wider variety of microphotographs from patients with leprosy and healthy skin, and the inclusion of other differential diagnoses of leprosy. This is crucial to improve the model's adaptability to different clinical contexts.

In conclusion, ANN models emerge as promising tools in leprosy detection due to their ability to identify positive cases with precision and sensitivity. Their capability in detecting early signs of the disease can facilitate faster and more accurate diagnosis, thereby improving clinical management and timely treatment of patients. Despite their strengths, it is crucial to continue exploring and refining these models to further enhance their performance and adaptability in diverse clinical settings. These advancements could significantly transform histopathological evaluation by offering more effective and accessible tools for diagnosing complex dermatological diseases such as leprosy.

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